

Lecture 8: First fundamental form (cont), surface integrals, vector fields

An **orientation** of a two-dimensional vector space is a choice of which ordered bases are positive: two bases have the same orientation when their change-of-basis matrix has positive determinant. Exchanging the two basis vectors reverses the orientation.

A surface is **orientable** if we can choose orientations of all its tangent planes consistently as the point varies. Precisely, we require local parametrizations covering S whose coordinate changes $\tilde{f} = f \circ \phi$ satisfy

$$\det D\phi > 0 \quad \text{on every overlap.}$$

Then the ordered bases (f_u, f_v) give compatible orientations. An **oriented surface** is an orientable surface together with a particular choice of orientation.

Every surface patch has a smooth unit normal field

$$N_f(f(u, v)) = \frac{f_u \times f_v}{\|f_u \times f_v\|}.$$

The denominator is nonzero because f is an immersion. Reversing the order of the parameters changes N_f to $-N_f$.

The issue is global: can these local choices be made to agree on all overlaps? Local unit normals always exist; a global one need not.

Coordinate changes on overlaps. Let $f : U \rightarrow S$ and $\tilde{f} : \tilde{U} \rightarrow S$ be embedded parametrizations, with $U, \tilde{U} \subset \mathbb{R}^2$ open. Their images overlap in the surface patch

$$O = f(U) \cap \tilde{f}(\tilde{U}).$$

The inverse $f^{-1} : f(U) \rightarrow U$ recovers the parameters of a surface point. It is *not* an inverse defined on all of $\mathbb{R}^3$.

Restrict to the parameter sets corresponding to the overlap:

$$U_0 = f^{-1}(O), \quad \tilde{U}_0 = \tilde{f}^{-1}(O).$$

The *coordinate change* is

$$\phi = f^{-1} \circ \tilde{f} : \tilde{U}_0 \longrightarrow U_0.$$

Read this composition from right to left:

$$\underbrace{(r, s)}_{\text{parameters for } \tilde{f}} \xrightarrow{\tilde{f}} \underbrace{p}_{\text{point of } O} \xrightarrow{f^{-1}} \underbrace{(u, v)}_{\text{parameters for } f}.$$

Thus $\phi(r, s) = (u, v)$ means that

$$\tilde{f}(r, s) = f(u, v), \quad \tilde{f} = f \circ \phi \quad \text{on } \tilde{U}_0.$$

The surface point does not move; only its coordinates change. The map ϕ is a diffeomorphism between open subsets of $\mathbb{R}^2$, with inverse $\tilde{f}^{-1} \circ f$. Hence $\det D\phi \neq 0$; its sign compares the two coordinate orientations.

Example: a negative transition determinant. On the plane $S = \{z = 0\}$, take two embeddings with domain $U = (-2, 2) \times (-1, 1)$:

$$f(u, v) = (u, v, 0), \quad \tilde{f}(r, s) = (s, r, 0).$$

Their images are horizontal and vertical rectangles. They overlap in

$$f(U) \cap \tilde{f}(U) = \{(x, y, 0) : |x| < 1, |y| < 1\}.$$

On the corresponding parameter square $(-1, 1)^2$,

$$\phi = f^{-1} \circ \tilde{f}, \quad \phi(r, s) = (s, r), \quad \tilde{f} = f \circ \phi.$$

Thus

$$D\phi = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \det D\phi = -1.$$

The ordered coordinate bases give opposite unit normals:

$$f_u \times f_v = (0, 0, 1), \quad \tilde{f}_r \times \tilde{f}_s = (0, 0, -1).$$

These two parametrizations are not orientation-compatible. This does *not* mean the plane is nonorientable: reversing the order of the parameters in the second patch makes the normals agree. Orientability asks whether compatible choices *can* be made.

Example: a positive transition determinant on the sphere. Let $D = \{(a, b) : a^2 + b^2 < 1\}$. The embeddings

$$\begin{aligned} f(u, v) &= (u, v, \sqrt{1 - u^2 - v^2}), \\ \tilde{f}(r, s) &= (\sqrt{1 - r^2 - s^2}, r, s) \end{aligned}$$

parametrize the upper hemisphere $z > 0$ and the right hemisphere $x > 0$, respectively. Coordinate projections give their inverses. Their images overlap where $x > 0$ and $z > 0$.

In the second parameter domain the overlap is $s > 0$. Set $x = \sqrt{1 - r^2 - s^2} > 0$. Then

$$\phi = f^{-1} \circ \tilde{f}, \quad \phi(r, s) = (\sqrt{1 - r^2 - s^2}, r) = (x, r).$$

Indeed, $f(\phi(r, s)) = (x, r, \sqrt{s^2}) = (x, r, s)$, since $s > 0$. Differentiating,

$$D\phi = \begin{pmatrix} -r/x & -s/x \\ 1 & 0 \end{pmatrix}, \quad \det D\phi = \frac{s}{x} > 0.$$

The orientations agree: at the same point $p = (x, y, z)$,

$$f_u \times f_v = \frac{p}{z}, \quad \tilde{f}_r \times \tilde{f}_s = \frac{p}{x}.$$

Both unit normals are the outward normal $N(p) = p$.

Theorem. A smooth surface $S \subset \mathbb{R}^3$ is orientable if and only if it admits a smooth unit normal field $N : S \rightarrow \mathbb{R}^3$.

Proof sketch. If $\tilde{f}(r, s) = f(\phi(r, s))$, the chain rule and bilinearity of the cross product give, at corresponding points,

$$\tilde{f}_r \times \tilde{f}_s = \det D\phi (f_u \times f_v).$$

Consequently the associated local unit normals satisfy

$$N_{\tilde{f}} = \frac{\det D\phi}{|\det D\phi|} N_f.$$

For an oriented collection of patches, all these determinants are positive. The local unit normals therefore agree on overlaps and define a smooth unit normal field on S .

Conversely, given N , declare a basis (v, w) of $T_p S$ positive when

$$\langle v \times w, N(p) \rangle > 0.$$

On each sufficiently small connected patch, $N_f = N$ or $N_f = -N$. If necessary, exchange its parameters to obtain $N_f = N$. The displayed transformation law then forces all transition determinants to be positive. Thus S is orientable.

A continuous unit normal field suffices: locally it equals $\pm N_f$, with a locally constant sign, so it is automatically smooth.

On the sphere S_R^2 , the fields

$$N(p) = \frac{p}{R} \quad \text{and} \quad -N(p) = -\frac{p}{R}$$

give the outward and inward orientations. For the cylinder $x^2 + y^2 = R^2$, a global unit normal is

$$N(x, y, z) = \frac{(x, y, 0)}{R}.$$

More generally, if a surface is a regular level set of a *single* smooth function on an open subset of $\mathbb{R}^3$,

$$S = F^{-1}(0), \quad \nabla F \neq 0 \text{ on } S,$$

then it is orientable. Indeed, $T_p S = \ker dF_p$, so

$$N(p) = \frac{\nabla F(p)}{\|\nabla F(p)\|}$$

is a global smooth unit normal field. Separate local defining functions do not by themselves give a consistent global normal.

On a connected orientable surface there are exactly two choices of unit normal, N and $-N$: any other choice equals $\varepsilon(p)N(p)$, where the continuous function $\varepsilon : S \rightarrow \{1, -1\}$ must be constant.

Orientability concerns a *normal* field, not a nowhere-vanishing tangent field. We can now choose a unit normal and study how it changes along the surface; this will lead to the second fundamental form.

Example: the Möbius strip is nonorientable. Consider the surface parametrized locally by

$$f(\theta, t) = \begin{pmatrix} (1 + t \cos(\theta/2)) \cos \theta \\ (1 + t \cos(\theta/2)) \sin \theta \\ t \sin(\theta/2) \end{pmatrix}, \quad |t| < \frac{1}{2}.$$

As θ runs around the circle, the transverse direction makes a half-turn. In particular,

$$f(\theta + 2\pi, t) = f(\theta, -t).$$

The image is a Möbius strip with its boundary omitted, so it is a surface without boundary. Small parameter rectangles give embedded patches; f is not one globally injective parametrization.

At the seam, the parameters are identified by

$$(\theta, t) \longmapsto (\theta + 2\pi, -t), \quad D\phi = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

This change of coordinates is invertible, but has determinant -1 . Unlike the plane example, its sign cannot be corrected consistently around the entire strip.

A negative determinant alone does not prove that claim. We will show that *any* continuous unit normal field would give a contradiction along the central circle

$$c(\theta) = f(\theta, 0) = (\cos \theta, \sin \theta, 0).$$

Along the central circle of the Möbius strip,

$$\begin{aligned}f_{\theta}(\theta, 0) &= (-\sin \theta, \cos \theta, 0), \\f_t(\theta, 0) &= (\cos(\theta/2) \cos \theta, \cos(\theta/2) \sin \theta, \sin(\theta/2)).\end{aligned}$$

These vectors are orthonormal. Their cross product is therefore a unit normal along the parameter interval:

$$n(\theta) = \begin{pmatrix} \sin(\theta/2) \cos \theta \\ \sin(\theta/2) \sin \theta \\ -\cos(\theta/2) \end{pmatrix}.$$

After one circuit,

$$c(2\pi) = c(0), \quad n(2\pi) = -n(0).$$

Thus n does not define a single-valued normal on the circle.

Suppose a continuous unit normal field N existed on the strip. At each point of the central circle it must satisfy

$$N(c(\theta)) = \varepsilon(\theta)n(\theta), \quad \varepsilon(\theta) \in \{1, -1\}.$$

Since $\varepsilon(\theta) = \langle N(c(\theta)), n(\theta) \rangle$ is continuous on $[0, 2\pi]$, it is constant. Hence

$$N(c(2\pi)) = -N(c(0)),$$

although $c(2\pi) = c(0)$. This is impossible for a unit vector. There is no global unit normal field, so the Möbius strip is *nonorientable*.